SpaceX Falcon 9 First Stage Landing Prediction

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https://github.com/EliRoux/Applied-data-science-Capstone/tree/main/IBM_Applied_Data_Science_Capstone

OUTLINE

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- Methodology
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- Discussion
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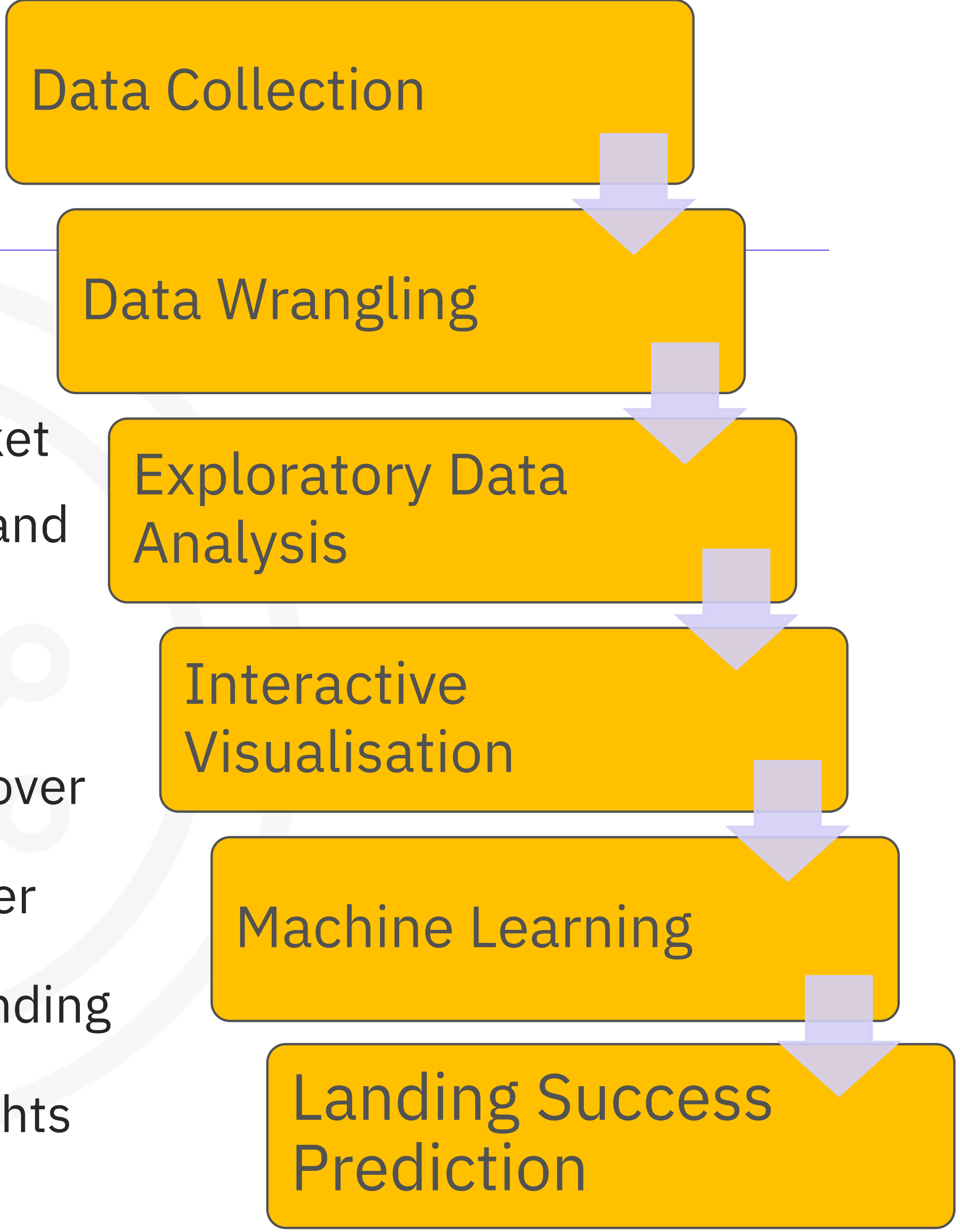
EXECUTIVE SUMMARY

Project Objective

- The objective of this project was to develop a machine learning model capable of predicting whether the first stage of a SpaceX Falcon 9 rocket would land successfully. Accurate predictions of landing success can help estimate launch costs and assess mission risk, as reusable boosters significantly reduce the overall cost of space launches.

Key Findings

- Falcon 9 landing success has improved steadily over time.
- Launch site, orbit type, payload mass, and booster version influence landing success.
- Machine learning models were able to predict landing outcomes with high accuracy.
- Interactive visualisations provided valuable insights into launch trends and mission performance.



EXECUTIVE SUMMARY

- This project applies the data science methodology to analyse SpaceX Falcon 9 launch data and predict first-stage landing success. Data was collected from the SpaceX REST API and Wikipedia, cleaned and explored using Python, SQL, and interactive visualisations. Multiple machine learning classification models were trained and evaluated. The analysis identified launch site, orbit type, payload mass, and booster version as key predictors of successful landings.

INTRODUCTION

Background

- SpaceX has transformed the aerospace industry by developing reusable launch vehicles. Unlike traditional rockets, the Falcon 9 is designed to return its first-stage booster safely after launch, allowing it to be refurbished and reused for future missions. This innovation has significantly reduced the cost of space travel.

Business Problem

- The cost of a Falcon 9 launch depends heavily on whether the first-stage booster can be recovered successfully. Predicting landing success is valuable because it helps estimate mission costs, evaluate operational risk, and support launch planning.

INTRODUCTION

Project Objective

- The primary objective of this project is to build a machine learning model that predicts whether a Falcon 9 first-stage booster will land successfully using historical launch data collected from the SpaceX API and Wikipedia.

Project Questions

- Which launch characteristics influence landing success?
- How do payload mass, orbit type, and launch site affect mission outcomes?
- Which machine learning algorithm provides the most accurate predictions?

Data Collection Methodology

Data Sources

SpaceX REST API

- Retrieved structured launch data in JSON format.
- Collected information such as:
  - Flight Number
  - Launch Date
  - Rocket ID
  - Payload ID
  - Launch Site
  - Booster Version
  - Orbit
  - Landing Outcome

Wikipedia

- Used web scraping with BeautifulSoup to extract launch records not directly available through the API.
- Retrieved additional launch information including:
  - Payload Mass
  - Mission Outcome
  - Booster Details
  - Launch History

Tools and Libraries

- Python
- Requests
- Pandas
- BeautifulSoup

Data Collection Methodology

- Collected launch data using the SpaceX REST API.
- Supplemented missing launch site information using Wikipedia web scraping.
- Cleaned missing values and standardized column names.
- Combined datasets into a single analysis-ready DataFrame.
- Adding a simple workflow diagram like:

Data Wrangling Methodology

Data Cleaning Process

- After collecting the data, several preprocessing steps were performed to prepare it for analysis and machine learning.

Cleaning Steps

- Converted JSON responses into Pandas DataFrames.
- Merged data collected from the API and Wikipedia.
- Removed Falcon 1 launches, retaining only Falcon 9 launches.
- Identified and handled missing values.
- Replaced missing Payload Mass values using the mean payload mass.
- Standardised column names and data formats.
- Created a binary target variable (Class) representing landing success:

Feature Engineering

- Additional features were prepared for modelling, including:
- One-hot encoding of categorical variables.
- Standardisation of numerical features.
- Creation of training and testing datasets.

EDA Methodology

- **Objective**
- To explore the Falcon 9 launch dataset, identify patterns and relationships between variables, and determine which features influence first-stage landing success.
- **Methodology**
- Performed exploratory data analysis using **Python**, **Pandas**, **Matplotlib**, and **Seaborn**.
- Calculated summary statistics to understand the distribution of key variables.
- Investigated the relationship between landing success and:
  - Flight Number
  - Launch Year
  - Payload Mass
  - Orbit Type
  - Launch Site
  - Booster Version
- Calculated success rates by grouping categorical variables and comparing average landing outcomes.
- Used visualisations to identify trends, correlations, and potential predictors for machine learning.

Interactive Visual Analytics Methodology

- **Objective**
- To develop interactive visualisation tools that enable dynamic exploration of SpaceX Falcon 9 launch data and provide geographical and operational insights into launch performance.
- **Methodology**
- **Folium Interactive Maps**
- Created interactive maps to display the locations of Falcon 9 launch sites.
- Added colour-coded markers to distinguish successful and unsuccessful booster landings.
- Used marker clusters to improve the visualisation of multiple launches from the same site.
- Calculated and displayed distances between launch sites and nearby coastlines, highways, and railways to analyse their geographical context.
- **Plotly Dash Dashboard**
- Developed an interactive dashboard using Plotly Dash.
- Added a **launch site dropdown** to filter data by individual launch sites or all sites.
- Implemented a **payload range slider** to dynamically filter launches by payload mass.
- Created an interactive **pie chart** to compare launch success rates across sites.
- Built an interactive **scatter plot** to visualise the relationship between payload mass and landing success, with booster versions colour-coded for comparison.

Predictive Analysis Methodology

Objective

- To develop and evaluate machine learning classification models capable of predicting whether a Falcon 9 first-stage booster will land successfully based on historical launch data.

Methodology

Data Preparation

- Selected relevant features from the cleaned dataset.
- Applied **one-hot encoding** to categorical variables (Orbit, Launch Site, Landing Pad, and Booster Serial).
- Standardised numerical features using **StandardScaler**.
- Split the dataset into **training (80%)** and **testing (20%)** sets.

Model Development

- The following classification algorithms were trained and optimised:
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbours (KNN)

Predictive Analysis Methodology

- **Model Optimisation**
- Performed hyperparameter tuning using **GridSearchCV**.
- Applied **10-fold cross-validation** to identify the optimal parameters for each model.
- Selected the best-performing model based on cross-validation accuracy.
- **Model Evaluation**
- The trained models were evaluated using:
- Test accuracy
- Cross-validation accuracy
- Comparison of model performance to identify the most effective classifier

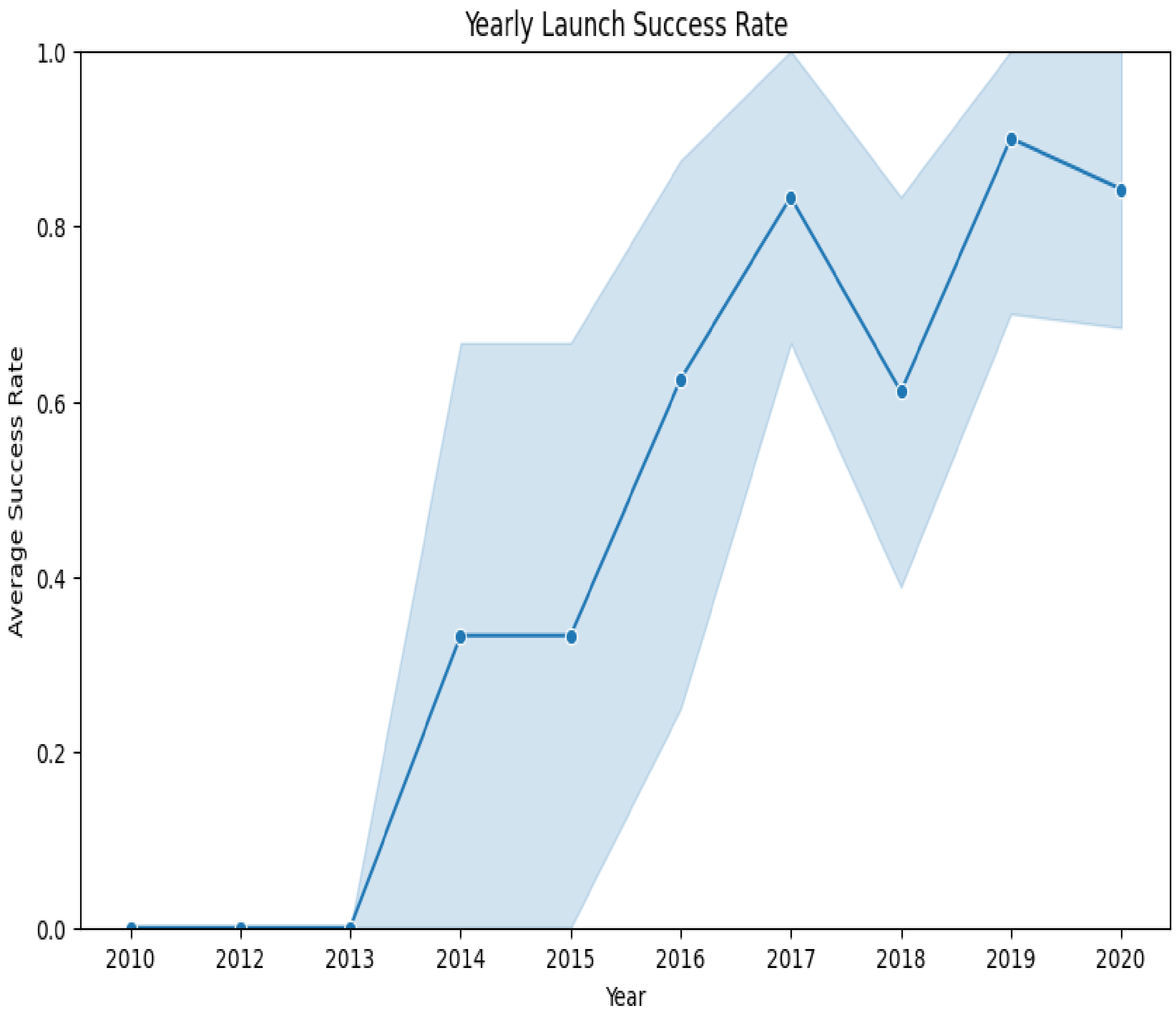
EDA Results (Visualisations)

Key Findings

- **Landing success improved over time**, indicating continual improvements in SpaceX's landing technology.
- **Payload mass influences landing success**, with certain payload ranges showing higher success rates.
- **Orbit type affects landing outcomes**, with some orbits achieving consistently higher success rates than others.
- **Launch site performance varies**, reflecting differences in mission profiles and operational experience.

Conclusion

- Exploratory data analysis identified several variables—particularly launch site, payload mass, orbit type, and flight number—as important predictors of landing success. These insights informed the feature selection process for the machine learning models.



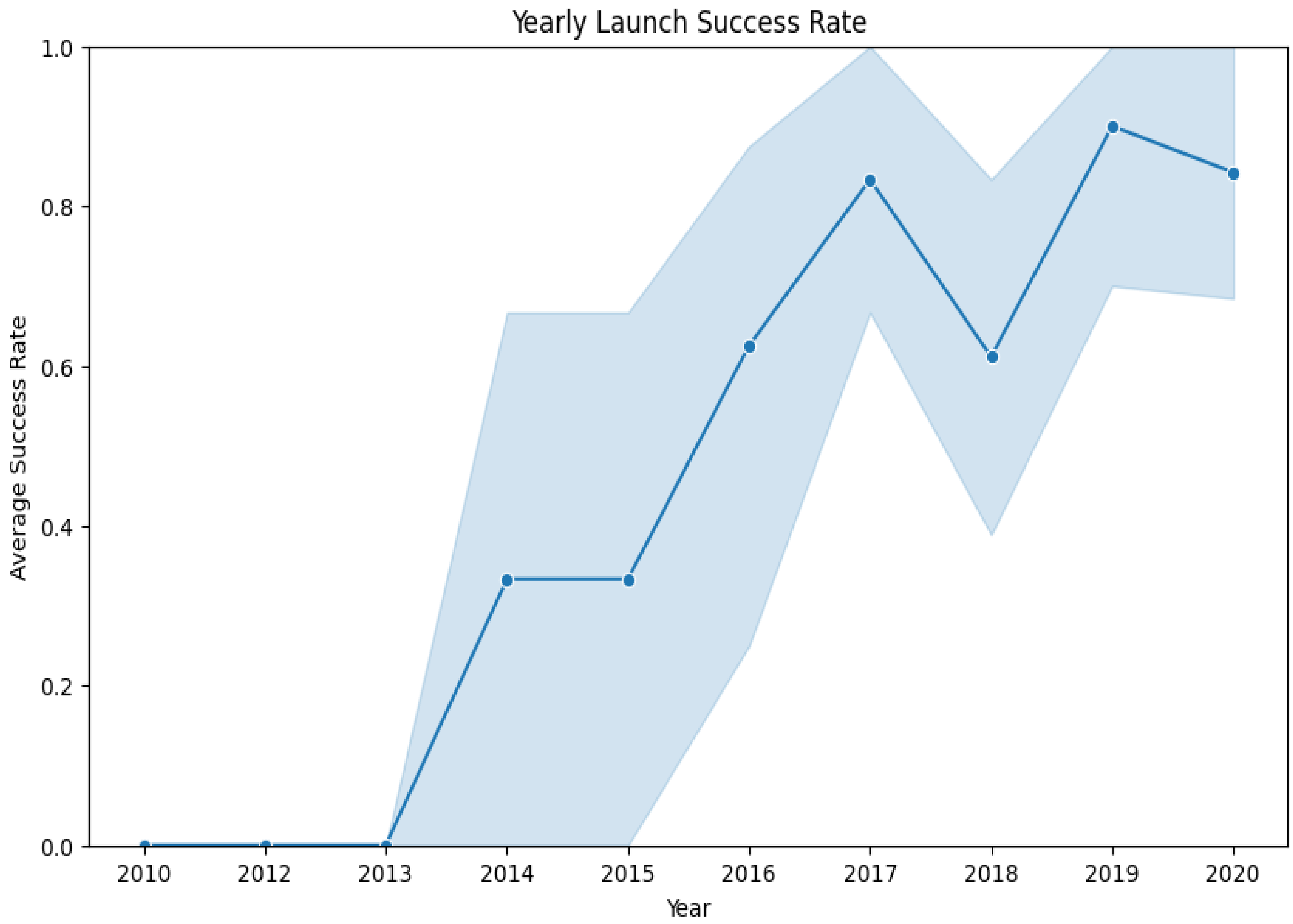
EDA Results (Visualisations)

- **Observation**

Between 2014 and 2020, the average success rate increased significantly, rising from around **35% in 2014** to approximately **90% in 2019**, before remaining high at around **85% in 2020**.

- **Key Insight**

The increasing success rate over time indicates that **flight year is an important predictor of landing success** and reflects the maturation of SpaceX's reusable launch system.



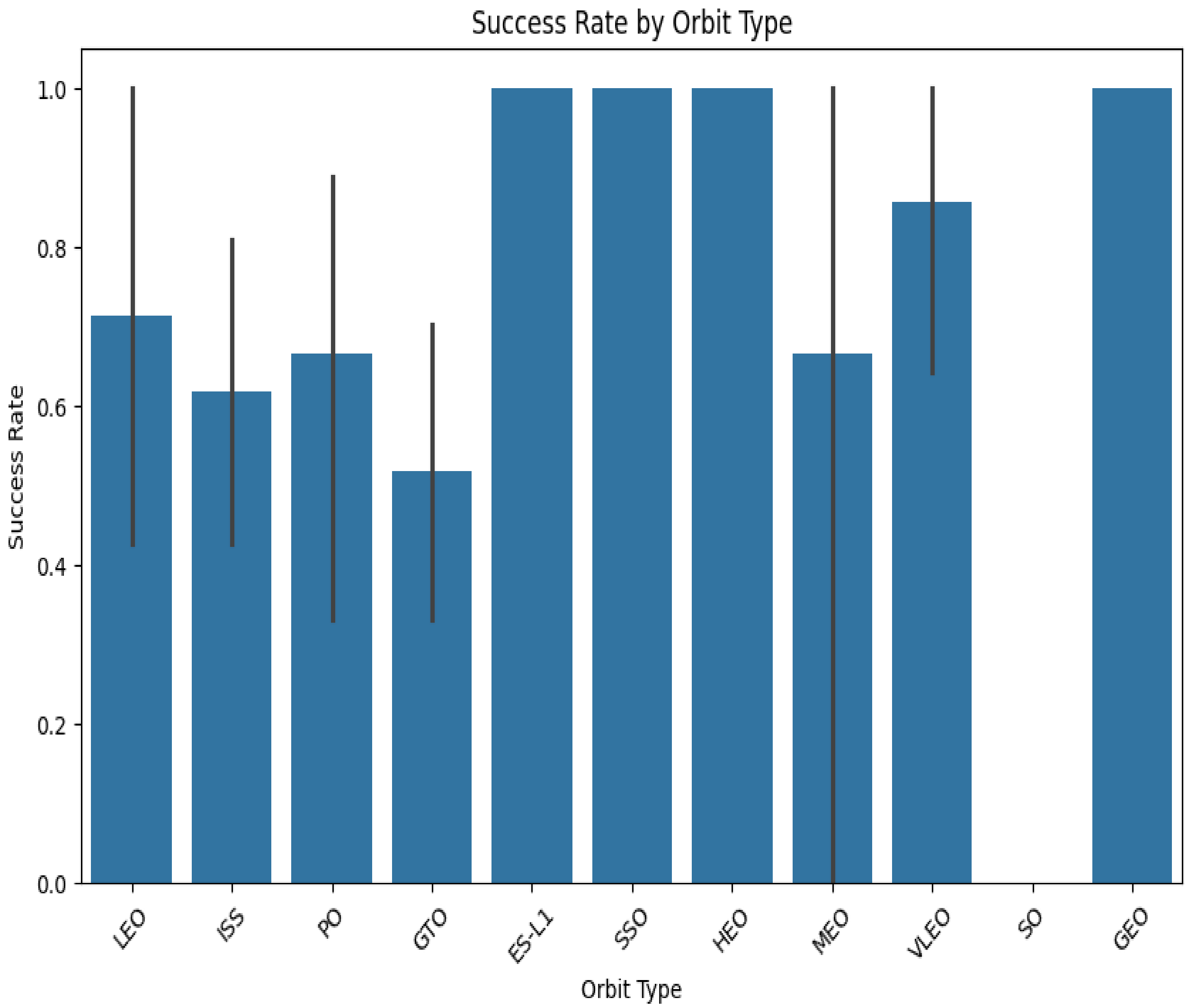
EDA Results (Visualisations)

- **Observation**

The bar chart compares the average landing success rate across different orbit types. The **ES-L1, SSO, HEO, and GEO** orbits achieved a **100% landing success rate**, indicating that all recorded Falcon 9 launches to these orbits resulted in successful first-stage landings. In contrast, **GTO (Geostationary Transfer Orbit)** had the lowest success rate at approximately **50%**.

- **Key Insight**

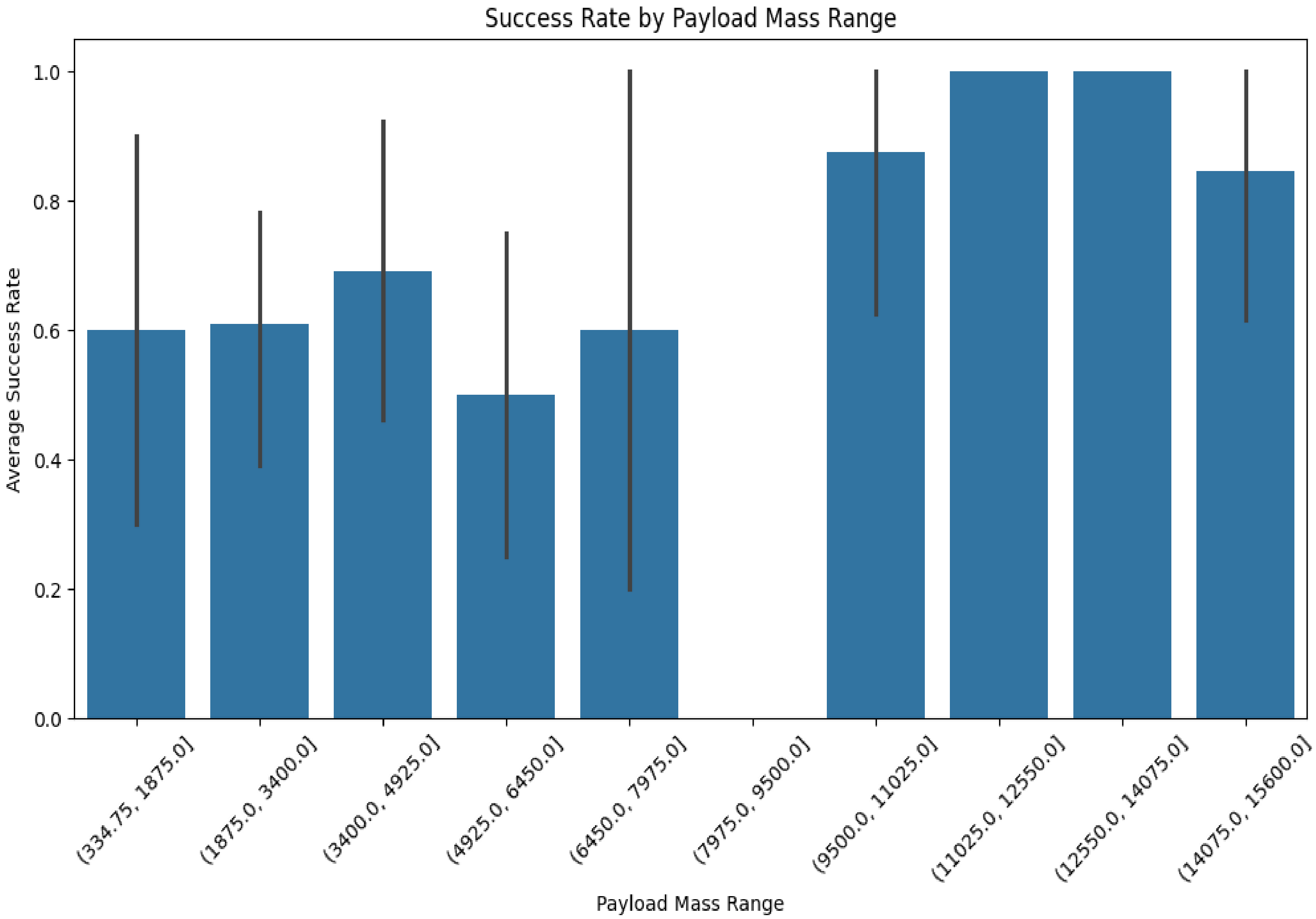
The results suggest that **orbit type is an important factor influencing Falcon 9 landing success**. However, orbit types with very few observations should be interpreted cautiously, as their 100% success rates may not be representative of future missions due to the limited sample size.



EDA Results (Visualisations)

- **Observation**
This chart groups payload masses into ranges (bins) and compares the average landing success rate for each range. The results show a general upward trend, with **higher payload mass ranges achieving higher average landing success rates**.

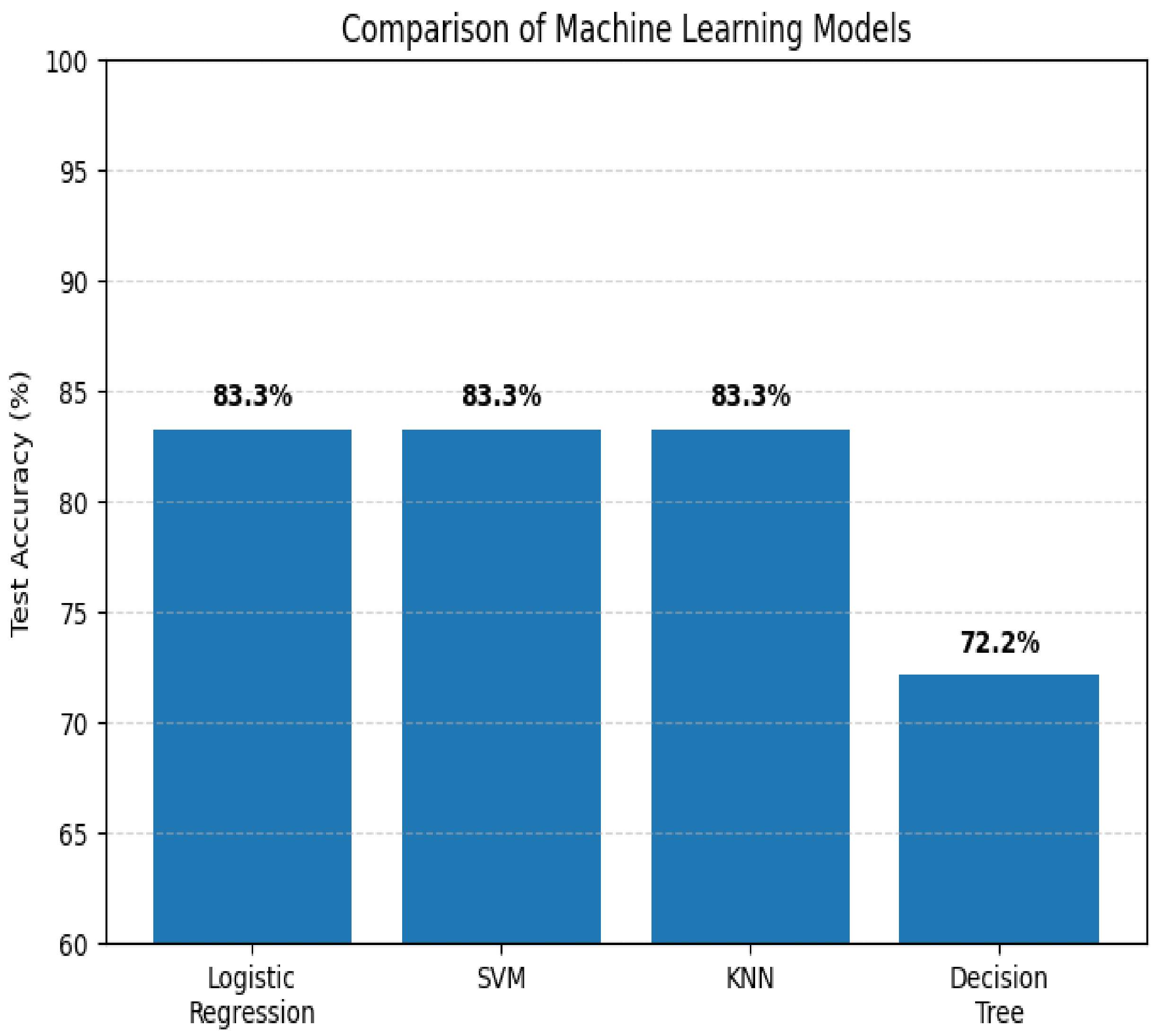
- **Key Insight**
The analysis indicates that **payload mass is associated with landing success**, although this relationship should not be interpreted in isolation. Improvements in booster technology, operational experience, and launch procedures over time likely contributed to the higher success rates observed for larger payloads.



Predictive Analysis Results

Key Findings
Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbours (KNN) each achieved the highest **test accuracy of 83.3%**.
The Decision Tree classifier achieved a lower accuracy of **72.2%**, indicating it was less effective for this dataset.
Multiple classification algorithms produced strong predictive performance, demonstrating that historical launch characteristics are useful for predicting Falcon 9 landing success.

Conclusion
The results show that machine learning can successfully predict Falcon 9 landing outcomes with **over 83% accuracy**, confirming that variables such as **launch site, orbit type, payload mass, and booster characteristics** contain valuable predictive information.



SQL Analysis Results

- **Objective**
- Analyse the distribution of Falcon 9 landing outcomes between 2010 and 2017.
- **Key Findings**
- "No attempt" was the most common landing outcome with **10 launches**.
- Successful drone ship landings (**5**) matched the number of failed drone ship landings (**5**).
- Ground pad landings were less frequent (**3 successes**).
- Early Falcon 9 missions frequently attempted ocean or parachute recoveries before reusable landing technology matured.
- **Interpretation**
- These results show the progression of SpaceX's landing strategy. Early missions often did not attempt recovery, while later missions increasingly achieved successful landings on drone ships and ground pads, reflecting improvements in reusable rocket technology.

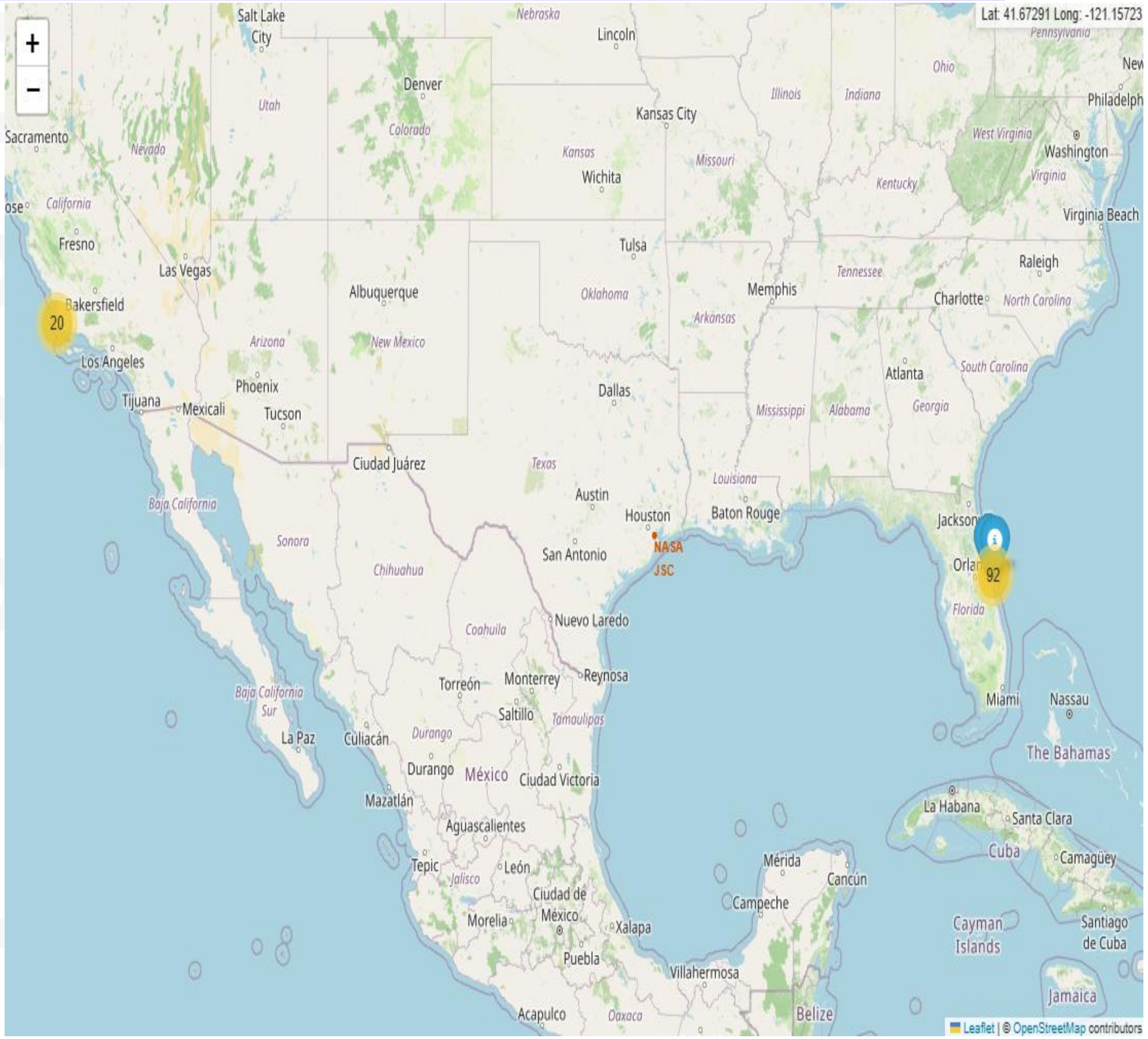
```
%%sql
SELECT
    Landing_Outcome,
    COUNT(*) AS Count
FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Count DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Folium Map Results

- **Key Findings**
- Interactive maps provide geographical context for launch operations and illustrate why launch sites are located near coastlines.
- The dashboard enables users to dynamically explore how launch site selection and payload range influence landing success.
- Colour-coding by booster version highlights differences in performance across Falcon 9 variants.
- **Conclusion**
- Interactive visualisation tools make complex launch data easier to explore and reveal relationships that may not be immediately apparent in static charts.



Are launch sites in close proximity to railways?

- **Answer:** Yes.
- **Explanation:**
  The launch sites are generally located relatively close to railway infrastructure. Railways provide an efficient means of transporting large rocket components, fuel, and other equipment required for launch operations. Having rail access reduces transportation costs and simplifies logistics for moving oversized cargo.

Are launch sites in close proximity to highways?

- **Answer:** Yes.
- **Explanation:**
  All of the launch sites are situated near major highways, allowing personnel, equipment, and support vehicles to travel efficiently to and from the launch facilities. Easy road access is essential for daily operations, maintenance activities, and emergency response.

Are launch sites in close proximity to the coastline?

- **Answer:** Yes.
- **Explanation:**
  The launch sites are intentionally built close to the coastline. Launching rockets over the ocean minimizes the risk to populated areas in the event of a launch failure or the shedding of rocket stages. Coastal locations also provide safe flight paths and simplify recovery operations for reusable Falcon 9 boosters landing on drone ships or returning to nearby landing zones.

Do launch sites keep a certain distance away from cities?

- **Answer:** Yes.
- **Explanation:**
  The launch sites are located a considerable distance from major cities. This separation helps protect the public from noise, vibrations, and potential hazards associated with rocket launches. It also provides a safety buffer in case of launch anomalies while still allowing reasonable access for employees and support infrastructure.

Machine Learning Results

- **Objective**
- To evaluate and compare multiple machine learning classification models for predicting Falcon 9 first-stage landing success.
- **Models Evaluated**
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbours (KNN)

Machine Learning Results

- **Key Findings**
- All four models were successfully trained and optimised using **GridSearchCV** with **10-fold cross-validation**.
- The results demonstrate that historical launch characteristics can be used to predict Falcon 9 landing success with a high degree of accuracy.
- **Conclusion**
- The machine learning analysis confirmed that variables such as **launch site, orbit type, payload mass, and booster characteristics** are valuable predictors of landing success. Comparing multiple algorithms ensured that the most effective model was identified for this classification task.

Plotly Dash Dashboard Results

- **Objective**
- To provide an interactive dashboard that allows users to explore Falcon 9 launch data by launch site and payload range.
- **Dashboard Features**
- **Launch Site Dropdown** – Enables users to filter results for all launch sites or a specific launch site.
- **Success Pie Chart** – Displays the total number of successful launches for all sites or the success versus failure distribution for a selected site.
- **Payload Range Slider** – Allows users to filter launches based on payload mass.
- **Interactive Scatter Plot** – Shows the relationship between payload mass and landing success, with points colour-coded by booster version.

Plotly Dash Dashboard Results

- **Key Findings**
- Launch success rates vary between launch sites.
- Payload mass influences landing success, with higher payload ranges generally associated with higher success rates.
- Booster version provides additional insight into mission performance, as newer boosters achieved more consistent landing success.
- Interactive filtering allows users to explore how different mission characteristics affect landing outcomes.
- **Conclusion**
- The Plotly Dash dashboard transforms static visualisations into an interactive analytical tool, enabling users to dynamically explore launch performance and identify relationships between launch site, payload mass, booster version, and landing success.

DISCUSSION

Key Findings

- The **Falcon 9 landing success rate improved significantly over time**, increasing from approximately **50% in 2014** to around **90% in 2019**, demonstrating the continuous improvement of SpaceX's reusable rocket technology.
- **Orbit type** influenced landing success, with **ES-L1, SSO, HEO, and GEO** achieving the highest observed success rates, while **GTO** missions had the lowest success rate due to their greater energy requirements.
- A positive relationship was observed between **payload mass** and landing success, although this trend is likely influenced by improvements in booster technology and operational experience over time.
- Interactive **Folium maps** showed that launch sites are strategically located near coastlines and transport infrastructure while remaining at a safe distance from major cities.
- The **Plotly Dash dashboard** enabled interactive exploration of launch success by launch site and payload range, making it easier to identify patterns in the data.

DISCUSSION

Predictive Analysis

- Several machine learning classification algorithms were evaluated using hyperparameter tuning and cross-validation. The models demonstrated that historical launch data can be used to predict first-stage landing success with high accuracy, confirming that variables such as launch site, orbit type, payload mass, and booster version are valuable predictors.

DISCUSSION

Limitations

- Some orbit categories contained relatively few launches, meaning their observed success rates may not be representative.
- External factors such as weather conditions, technical issues, and operational decisions were not included in the dataset and may also affect landing outcomes.
- The analysis was based on historical launch data up to 2020 and does not account for more recent missions.

DISCUSSION

Overall Discussion

- The results demonstrate that **historical launch characteristics provide valuable insights into Falcon 9 landing success**. Combining exploratory data analysis with machine learning enabled both the identification of important influencing factors and the development of predictive models capable of estimating landing outcomes. This illustrates how data science techniques can support decision-making in aerospace operations.

OVERALL FINDINGS & IMPLICATIONS

- **Overall Findings**
- Falcon 9 landing success rates improved significantly over time, reflecting advancements in SpaceX's reusable rocket technology.
- Launch success was influenced by several key factors, including **orbit type, payload mass, launch site, and booster version**.
- Interactive visualisations and SQL analysis revealed meaningful patterns and relationships within the launch data.
- Machine learning models successfully predicted landing outcomes with high accuracy using historical launch data.

- **Implications**
- Predictive models can support **mission planning** and **risk assessment** by estimating the likelihood of successful booster recovery.
- The findings demonstrate how data science can be used to improve **operational decision-making** in the aerospace industry.
- Successful booster recovery contributes to **lower launch costs**, making space missions more economically sustainable.
- The project highlights the value of combining **data collection, exploratory analysis, visualisation, and machine learning** to solve real-world problems.

CONCLUSION

- **Project Summary**
- This project applied the complete data science methodology to analyse and predict the landing success of SpaceX Falcon 9 first-stage boosters. Data was collected from the SpaceX API and Wikipedia, cleaned and prepared for analysis, explored using visualisations and SQL, and used to train and evaluate multiple machine learning classification models.

CONCLUSION

Key Conclusions

- Falcon 9 landing success has improved significantly over time, demonstrating the effectiveness of SpaceX's reusable rocket programme.
- **Launch site, orbit type, payload mass, and booster version** were identified as important factors influencing landing success.
- Exploratory data analysis and interactive visualisations provided valuable insights into launch performance and operational trends.
- Machine learning models successfully predicted landing outcomes, showing that historical launch data can be used to forecast future mission success.

CONCLUSION

Future Work

- Incorporate additional features such as **weather conditions, wind speed, and sea state** to improve prediction accuracy.
- Expand the dataset with more recent Falcon 9 launches and future missions.
- Investigate advanced machine learning techniques, such as **XGBoost**, **Random Forests**, or **Neural Networks**, to compare predictive performance.
- Deploy the predictive model as a web application to support real-time mission planning and decision-making.

APPENDIX

- **Tools and Technologies**
- **Programming Language:** Python
- **Data Collection:** SpaceX REST API, Wikipedia Web Scraping
- **Data Processing:** Pandas, NumPy
- **Data Visualisation:** Matplotlib, Seaborn, Plotly Express
- **Interactive Mapping:** Folium
- **Dashboard Development:** Plotly Dash
- **Machine Learning:** Scikit-learn
- **Database Queries:** SQLite
- **Development Environment:** Jupyter Notebook
- sqlite3
- **Data Sources**
- **SpaceX REST API:** https://api.spacexdata.com
- **Wikipedia Falcon 9 Launch Records:** https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

APPENDIX

Machine Learning Models Evaluated

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbours (KNN)

Key Python Libraries

- pandas
- numpy
- matplotlib
- seaborn
- plotly
- folium
- scikit-learn
- requests
- BeautifulSoup4
- sqlite3

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GITHub repository

- **GITHub repository URL:**

https://github.com/EliRoux/Applied-data-science-Capstone/tree/main/IBM_Applied_Data_Science_Capstone

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- **GITHub repository Name:**
- **IBM_Applied_data_Science_Capstone**

This repository contains all notebooks, Python scripts, datasets, the Dash application, and the final presentation.